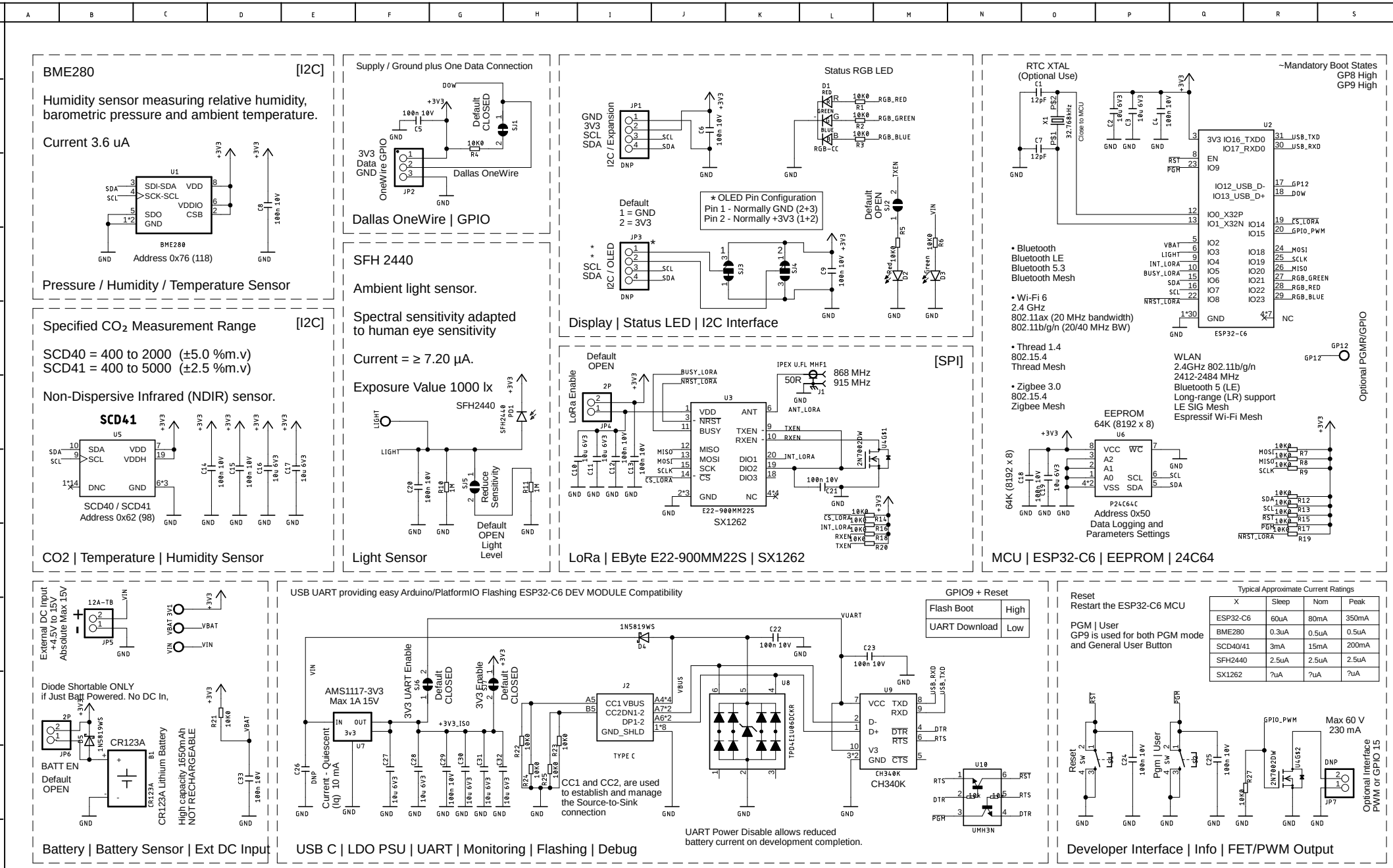
BME280
[I2C]
Humidity sensor measuring relative humidity, barometric pressure and ambient temperature.
Current 3.6 uA
Address 0x76 (118)
Pressure / Humidity / Temperature Sensor
Supply / Ground plus One Data Connection
Dallas OneWire | GPIO
SFH 2440
Ambient light sensor.
Spectral sensitivity adapted to human eye sensitivity
Current = ≥ 7.20 µA.
Exposure Value 1000 lx
Light Sensor
Status RGB LED
Display | Status LED | I2C Interface
RTC XTAL (Optional Use)
~Mandatory Boot States GP8 High GP9 High
• Bluetooth Bluetooth LE Bluetooth 5.3 Bluetooth Mesh
• Wi-Fi 6 2.4 GHz 802.11ax (20 MHz bandwidth) 802.11b/g/n (20/40 MHz BW)
• Thread 1.4 802.15.4 Thread Mesh
• Zigbee 3.0 802.15.4 Zigbee Mesh
WLAN 2.4GHz 802.11b/g/n 2412-2484 MHz Bluetooth 5 (LE) Long-range (LR) support LE SIG Mesh Espressif Wi-Fi Mesh
Optional PGMR/GPIO
EEPROM 64K (8192 x 8)
Address 0x50 Data Logging and Parameters Settings
MCU | ESP32-C6 | EEPROM | 24C64
Specified CO₂ Measurement Range
[I2C]
SCD40 = 400 to 2000 (±5.0 %m.v)
SCD41 = 400 to 5000 (±2.5 %m.v)
Non-Dispersive Infrared (NDIR) sensor.
SCD40 / SCD41 Address 0x62 (98)
CO2 | Temperature | Humidity Sensor
[SPI]
868 MHz 915 MHz
SX1262
LoRa | EByte E22-900MM22S | SX1262
External DC Input +4.5V to 15V Absolute Max 15V
Diode Shortable ONLY if Just Batt Powered. No DC In,
BATT EN Default OPEN
CR123A
High capacity 1650mAh NOT RECHARGEABLE
Battery | Battery Sensor | Ext DC Input
USB UART providing easy Arduino/PlatformIO Flashing ESP32-C6 DEV MODULE Compatibility
AMS1117-3V3 Max 1A 15V
Current - Quiescent (Iq) 10 mA
CC1 and CC2, are used to establish and manage the Source-to-Sink connection
CH340K
UART Power Disable allows reduced battery current on development completion.
USB C | LDO PSU | UART | Monitoring | Flashing | Debug
GPIO9 + Reset
Flash Boot High
UART Download Low
Reset Restart the ESP32-C6 MCU
PGM | User GP9 is used for both PGM mode and General User Button
Typical Approximate Current Ratings
Max 60 V 230 mA
Optional Interface PWM or GPIO 15
Developer Interface | Info | FET/PWM Output
DitroniX
Production
(C)2021-2026 DitroniX - Dave Williams G8PUO
Main PCB 63x51
ESP32-C5 SX1262 SDK
ditronix.net
DitroniX Ltd – UK
AQIL ESP32C6 LoRa SCD41 SDK v1.2603.102
AIR-QUALITY-MONITOR-IOT-WEATHER-STATION-LORA
Sheet 1/1
Drwg/Chk: Dave Williams
12/03/2026 16:17